

The true effect of AI on the environment.

Posted by u/Murky-Opposite6464 • • 0 points • 1 comments

I've written about this before, but I felt like this topic deserves a deeper, more granular analysis of the environmental impact of AI. What I intend to do is show the current environmental impact of AI, compare that to the improvements AI has proven it is capable of, and come up with an estimate of how much damage/benefit is done by AI overall.

Since I am pro-AI myself, I'm going to endeavour to be fair by going by the highest figures I can find, even if those are estimates of future use.

First, the easy part, environmental damage.

"Data centers' electricity consumption in 2026 is projected to reach 1,000 terawatts (tWh), roughly Japan's total consumption".

That combined with the fact that AI is estimated to soon be using 50% of data centre power leaves is with 500 terawatts

"AI demand for water in 2027 would be 4.2-6.6 billion cubic meters, roughly equivalent to the consumption of 'half of the United Kingdom'"

<https://begin.berkeley.edu/reducing-ais-climate-impact-everything-you-always-wanted-to-know-but-were-afraid-to-ask/>

I've excluded CO₂ emissions here because they are essentially a byproduct of energy use. Since the carbon intensity of electricity generation varies across regions, I chose to focus directly on the raw energy numbers, with the understanding that reductions in energy use will translate to a reduction in CO₂.

So, here are the scores.

Electricity use: 500 terawatt hours

Water use: 6.6 billion cubic meters

These are the numbers I will be comparing positive environmental impact with. And since I started writing this section first, I'm just as excited to see what the results will be as you are. :)

I'm going to be going through this field by field, sector by sector, and totalling up the results I find. To ensure total fairness, any range I am given, I will be using the lowest estimate.

Let's start with agriculture.

We are currently using 4.3 trillion cubic meters of fresh water per year, with 70% of that going to agriculture.[1] That means we are using 3 trillion cubic meters just for growing food.

I found many reports on AI controlled irrigation techniques reducing water usage by up to 30%, increasing crop yields by 20% at the same time. However, these were in third world countries that likely aren't using the most efficient non-AI methods to begin with. For a fairer comparison, I looked to a test performed in Canada that resulted in between 6.4% and 22.8% in water reduction, and a 2.3% to 4.3% increase in yield. So, going with my lowest estimate policy, that's a reduction of 6.4%, plus a negligible amount for increased crop yields[2].

If we were to apply this system globally, that's a 192 billion cubic meter reduction in water use. It can reduce herbicide use too by using AI targeting, but we won't factor that in.

Well... we sure past that 6.6 billion cubic meters of water lost metric quickly.

Let's move on to the next sector. Data centres!

The data centres that power AI, and the site you are reading this on, have been a big topic as far as increases in electrical use. However, little is said about how AI can make those data centres more efficient.

Google claimed that by using their Deepmind AI, they cut the cost of cooling their data centres by "up to" 40%, 15% of the data centres total power use.

I saw many claims of similar numbers, usually around 30%. The lowest I could find is a reduction in cooling costs from between 14% and 21%[3]. So, let's go with 14%.

If Googles 40% claim lead to a 15% reduction overall (which seems to match other sources), a 10% reduction on cooling should be a 5.25% total reduction. So 52.5 terawatt hours.

Air conditioning!

When it comes to temperature control in general for places like offices and departments stores, I found cuts of up to 67% in various studies. The lowest I could find was 24.52%[4]. Since it had a variance of 10%, we'll go with 14%.

To get the air conditioning power use for only large buildings, I'm simply going to subtract residential air conditioning energy use from total air conditioning use.

We get 2,100 terawatt hours for total air conditioning[5], and residential air conditioning units are 1,200 terawatt hours[6], so 900 terawatt hours for more industrial air conditioning. That's a reduction of 126 terawatt hours.

Solar panel and wind farm optimization!

This one was fairly straightforward. AI can help position solar panels to increase energy production by 20% (I saw claims as high as 30%), and wind farm power output by 12%[7].

Solar produces 1,600 terawatt hours per year[8], leading to an increase of 320 terawatt hours. Wind generates 2,330 terawatt hours[9], so an increase of 279 terawatt hours.

So, even by the lowest metrics I could find, the increased efficiency for wind and solar farms on its own would basically offset AI's energy consumption entirely, making everything else I've cited an actual reduction in global energy use. All thanks to AI.

I hope this will counter some of the misinformation I've been seeing out there, and put some concerns to rest. There will be challenges that come with the growth of AI, a lot will change. However, as far as the environment is concerned, these changes will be for the better.

1: https://reports.weforum.org/docs/WEF_Water_Futures_Mobilizing_Multi_Stakeholder_Action_for_Resilience_2025.pdf

2: <https://arxiv.org/abs/2306.08715>

3: <https://arxiv.org/abs/2501.15085>

4: https://www.researchgate.net/figure/The-average-energy-savings-of-the-24-cases-and-the-maximum-energy-savings-achieved-by_fig6_331608687

5: <https://ourworldindata.org/air-conditioning-causes-around-greenhouse-gas-emissions-will-change-future>

6: <https://www.nature.com/articles/s41467-024-52028-8>

7: https://ojs.stanford.edu/ojs/index.php/intersect/article/view/3541?utm_source=chatgpt.com

8: <https://www.iea.org/energy-system/renewables/solar-pv>

9: <https://www.iea.org/energy-system/renewables/wind>

Fun little bonus facts. The positive effects of AI on carbon emissions in just 3 industries (meat and dairy, power, and light road vehicles) would completely negate all damage done by increased power consumption from expanding AI use.

<https://www.nature.com/articles/s44168-025-00252-3>

UPS optimized their deliveries with AI, saving 10 million gallons of fuel.

https://www.bestpractice.ai/ai-case-study-best-practice/ups_saves_over_10_million_gallons_of_fuel_and_up_to_%24400m_in_costs_annually_with_advanced_telematics_and_ana

Pittsburg used AI controlled stoplights to cut emissions from cars by 21%, and reduced total driving time by 26%.

<https://govlaunch.com/stories/pittsburgh-pa-reduces-traffic-congestion-with-ai>

Comments

u/ilovegoodfood • 1 points • 2025-09-23 23:09 UTC

This is a very nice comparison.

I'll be using this comment to show this to a few folks.

u/[deleted] • 1 points • 2025-09-24 00:05 UTC

[removed]

u/Amethystea • 2 points • 2025-09-23 00:10 UTC

Another point often neglected is that AI is often used to replace less efficient systems. So, as AI takes on more workload, those other systems and sectors would see an overall decrease in usage.

u/Superseaslug • 2 points • 2025-09-24 08:58 UTC

That and optimizing much larger energy users, like agriculture and global logistics.

u/Quirky-Complaint-839 • 5 points • 2025-09-22 11:58 UTC

People said that crypto space did nothing creative and wasted electricity. Now people say something that is creativity is a waste on resources and harmful.

They are valid concerns though. But it goes beyond just AI.